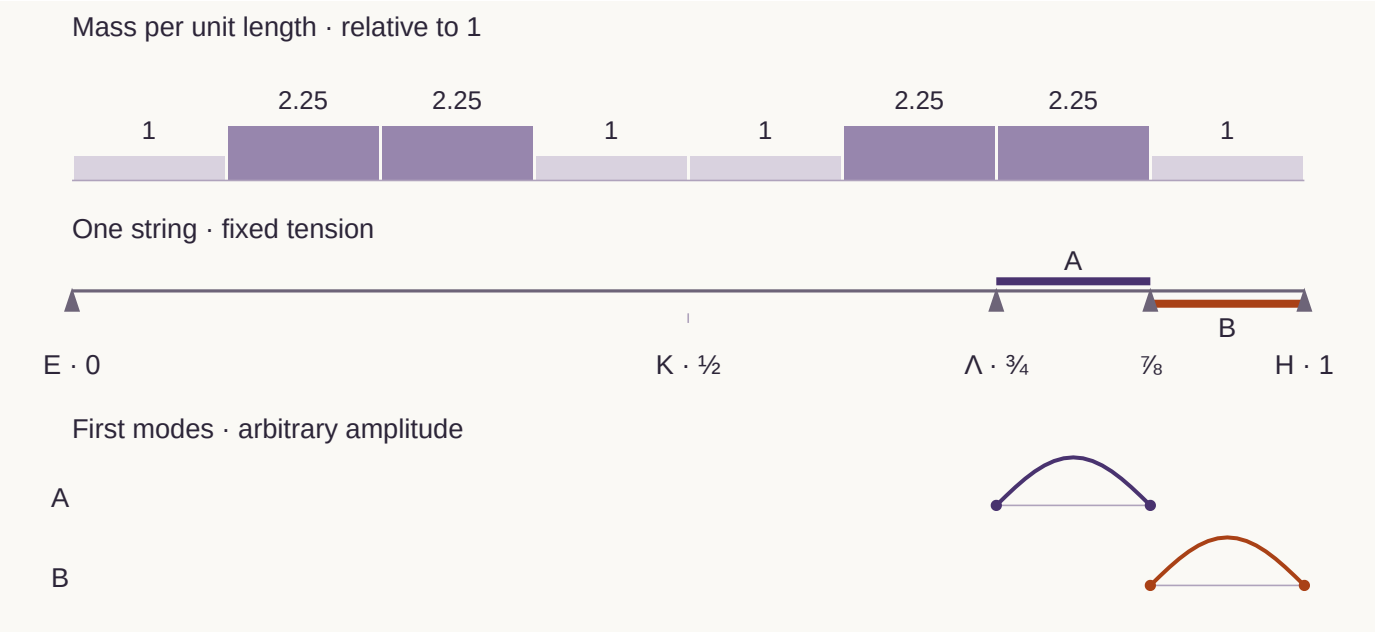
Equal lengths. Unequal notes.

Repeating hidden variation · heavy density 2.25 · uniform whole-string reference 220 Hz · selected test 3: One more test: eighths.



Eight chosen mass-density cells. The selected parts have equal length and fixed ideal tension. Triangles mark active supports; ticks mark unused ruler positions. First-mode amplitudes are arbitrary. Greek H (eta) corresponds to G in Wallis's Latin diagram; Λ to L.

These equal lengths sound different.

A: **1173.333333 Hz**; B: **1760.000000 Hz**. B minus A: 701.955001 cents. A computed agreement at one division does not prove uniform mass everywhere.

Embedded sound: A, then B. Generated sine fundamentals; no recovered ancient performance or timbre. The control duplicates A even when the selected B has a different frequency.

What each test establishes

All three comparisons · Repeating hidden variation, heavy density 2.25

Test	A · Hz	B · Hz	B − A · cents	Result
Halves	308.600377	308.600377	0.000000	Same pitch
Quarters	678.435281	678.435281	0.000000	Same pitch
Extra eighths	1173.333333	1760.000000	701.955001	Different pitch

The halves and right-hand quarters are Ptolemy’s comparisons. The eighths are our extra diagnostic. The uniform profile passes all three. Heavy outer quarters give matching halves but unequal right-hand quarters. A repeating mirrored pattern passes both ancient comparisons, but fails the extra eighths test. A heavy left half fails the first comparison while its light right-hand quarters agree.

Uniform segment: frequency = (1 / 2 length) × √(tension / mass per length).

Nonuniform segments require the wave equation through their actual mass distribution. This model uses eight piecewise-constant cells, ideal fixed boundaries and constant tension. A 110 or 220 Hz completely uniform full string sets the modern reference. It does not assign an ancient absolute pitch. The wave equation, mass profiles and additional eighths test are modern teaching choices. String stiffness, damping, bridge width, support compliance and changes of tension caused by a real bridge are omitted.

Equal pitch here means numerical agreement within 1e-7 cents, a computational tolerance rather than a human hearing threshold. Reflected mass distributions have matching spectra. Passing a finite set of comparisons cannot establish arbitrary material uniformity.

Read the examination

Ptolemy, *Harmonics* I.8, selected phrases from John Wallis’s Greek text and Latin translation (Oxford, 1682), pp. 35, 37–38. These are excerpts, not a continuous chapter translation. Greek ligatures and Unicode are normalized; segment letters are capitalized instead of barred lowercase. Latin long-s is normalized; & is retained. English translations and grammar notes are new.

1 · First, halve the whole

Πρῶτον μὲν ἐπὶ τὴν γινομένην τοῦ ὅλου μήκους διχοτομίαν

Primo quidem, ad factam totius longitudinis bisectionem

First, at the bisection of the whole length

An excerpt from the construction instruction. διχοτομίαν / bisectionem means division into two. τοῦ ὅλου μήκους / totius longitudinis is ‘of the whole length’: the genitive names what is being divided. The midpoint is K in Greek and K in Wallis’s Latin diagram. Printed p. 37.

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2 · Then halve the half

καὶ ἐπὶ τὴν τῆς ἡμισείας διχοτομίαν

deinde, ad semissis bisectionem

and at the bisection of the half

Another construction excerpt. The half from K to H is divided at Λ. The two new parts are quarters of the whole string. ἡμισείας / semissis means ‘of the half’. These are the source’s second comparison, rather than a second comparison of the original halves. Printed p. 37.

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3 · Equal in pitch?

τὸ ΕΚ τῆς χορδῆς μέρος, ἰσότονον εὑρίσκηται τῷ ΚΗ· καὶ ἔτι τὸ ΚΛ, τῷ ΛΗ

si chordae pars EK comperiatur ipsi KG aequitona; & KL, ipsi LG

if the string’s part EK is found equal in pitch to KH, and also KΛ to ΛH

Selected condition, not a complete sentence. ἰσότονον / aequitona means ‘equal in pitch’; it does not mean ‘equal in physical tension’. The print uses barred lowercase segment letters, normalized here to capitals. Greek H (eta) corresponds to Wallis’s Latin G, and Λ to L. The text continues by treating the string’s constitution as unvarying; the modern counterexample below explores the limit of that inference. Printed p. 37.

4 · When the trial fails

Ἐὰν δὲ μή· μεταφέρωμεν τὴν δοκιμασίαν ἐπ’ ἄλλο μέρος, ἤτοι χορδίῳ ἄλλῳ

sin minus; transferendam esse probationem ad aliam chordae partem, aut etiam ad chordam aliam

But if not, let us transfer the trial to another part, or to another string

An excerpt of the instruction following failure. ἐὰν δὲ μή / sin minus means ‘but if not’. μεταφέρωμεν is a ‘let us’ subjunctive; δοκιμασίαν / probationem is the trial or examination. Wallis casts the instruction as ‘the trial is to be transferred’. The instruction gives a concrete response to a failed instrument check. Printed p. 37.

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5 · What must stay comparable

τὸ ἐν τοῖς ὁμοίοις καὶ ἀναλόγοις καὶ ἰσομήκεσι καὶ μίαν ἔχουσι τάσιν, ὁμότονον

ut in similibus, proportionalibus, longitudine aequalibus, eandemque tensionem habentibus, sit tonus correspondens

sameness of pitch in like, corresponding, equal-length parts under one tension

The Greek clause crosses from p. 37 to p. 38. ἰσομήκεσι / longitudine aequalibus concerns equal lengths. μίαν ἔχουσι τάσιν / eandem tensionem habentibus concerns a single tension. ὁμότονον concerns pitch. Those are three different conditions. Our model fixes length and tension and varies mass per unit length. Printed p. 38.

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6 · An instrument for testing

Κανὼν ἁρμονικός

Canon Harmonicus

Harmonic canon

The heading of the instrument diagram on p. 35. κανών can mean a straight rod, rule or measuring instrument. The diagram is part of Wallis’s 1682 edition, not an ancient blueprint. His long note on p. 36 discusses and changes the drawn bridge surfaces; the complete pages retain that editorial evidence. Printed p. 35.

Three reading checks

The two halves agree. What follows?

These two measured pitches agree; try the quarters too. A mirrored mass profile gives the halves the same spectrum even when neither half is uniform.

What does ἰσότονον mean in this trial?

Equal in pitch. Ptolemy names pitch, length and tension separately. Agreement in one quantity does not establish the others.

The repeating profile passes both ancient comparisons. What does the eighths test show?

These two comparisons can miss a smaller pattern of variation. Eighths are our additional diagnostic. This chosen repeating profile is a counterexample to proving uniformity from only these two equal-pitch comparisons.

Reopen this experiment · Complete scanned edition · Derivation and source decisions.

Ptolemy’s instrument examination precedes his use of string divisions to test concords. The selected wording treats the string’s constitution as unvarying after agreement. The repeating modern example makes the limit of that inference explicit. Wallis discusses and changes the bridge drawing in his note on p. 36; the full scans retain it. No claim is made about the precision of a physical ancient instrument.

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τε, ϗ συμφώνων· οἱ δὲ τ τ λοιπῶν, ⓦ προσήκοντα διορισμὸν ἔξουσιν u ἐν τοῖς x οἰκείοις τόποις. Νῦν δὲ καλῶς ἂν ἔχοι τὴν y ἀναφορὰν ποιησάμενοι τῶν z ἤδη προβεβλημένων· ἵνα τὸ πρὸς τὴν αἴσθησιν αὐτῶν ὁμολογούμενον ἀδιαπτώτως ἔχωμεν a ὑποκείμενον.

perparticularium, duorumq; primorum Conſonorum:) De *Concinnis* vero reliquis; quænam ſint eorum rationes, commode definietur ſuis locis. Jam vero opportunum erit, evidentiam demonſtrare eorum quæ hactenus tradita ſunt: ut eorum cum ſenſu convenientiam indubitato ſubjectam habeamus.

ιʹ.

Τίνα τρόπον a ἀδιαπτώτως b δειχθήσονται τῆς συμφωνίας ⓐ λόγοι διὰ τοῦ μονοχόρδου κανόνος.

CAP. VIII.

Quo pacto Conſonantiarum Rationes indubitato oſtendantur, per monochordum Canonem.

Το μὲν οὖν ἀπὸ τῶν αὐλῶν, ϗ τῶν συρίγγων, πιθανὸν ὂν προσκείμενον, ἢ τῶν περὶ τὰς χορδὰς ἐξαρτωμένων βαρῶν, c ἀφείσθω. Διὰ τὸ μὴ δύνασθαι τὰς τοιαύτας d ἀποδείξεις ἐπὶ ὃ e ἀκριβέστατον φάναι· ἀλλὰ διαβολῆς μᾶλλον ἐμποιεῖν ἀφορμὰς τοῖς * πειρωμένοις. Ἐπί τε γὰρ τῶν αὐλῶν ϗ τῶν συρίγγων· μὴ ὅτι δυσεξέταστον αὐτῶν ἐστὶ τὸ ἀνωμαλίας διόρθωσιν· ἔτι ϗ τὰ πέρατα, πρὸς ἃ δεῖ τὰ μήκη προβάλλειν, οὐ πλάτει πως ὁρίζεται) f ἀόριστα· g ϗ διὰ ὅ, καθόλου, τοῖς πλείστοις τῶν ἐμπνευστῶν ὀργάνων, ἀταξίαν τινὰ προσγίνεσθαι, ϗ προβολὰς τοῦ πνεύματος ἐμβολάς. Ἐπὶ

PER Tibias igitur & Fiſtulas, exhibere quod eſt propoſitum; aut, per appenſa Chordis Pondera; omittamus: eo quod non poſſint hujuſmodi demonſtrationes ad ſummam accurationem perduci; ſed calumniandi potius occaſionem exhiberent, id conantibus. Nam in tibiis & fiſtulis; præterquam quod ſit admodum difficile omnem irregularitatem inibi cavere: etiam termini, ad quos ſunt exigendæ longitudines, latitudinem quandam admittunt indefinitam: atque (in univerſum) Inſtrumentorum inflatilium pleraque, inordinatum aliquid

τ AB. λοιποὶ. u C. deeſt ἐν. x S. τοπους οἰκείοις. y AB. ἐνεργ--. z S. ἴσιν. a ABS. ὑποπεμένον. CAP. 8. a AB. ἀδίας--. b GI. --σεται. c S. --σθω. d K. ἀποδ--. e GIS. --τερον. * S. βαλομένοις. f CGI, deeſt ἀορίστα. g CS. μὴ ὅτι ϗ κα--.

quid adjunctum habent; & præter ipſas ſpiritus injectiones. In appenſis autem ad chordas ponderibus; præterquam quod ipſæ chordæ haud facile conſerventur inter ſe penitus invariatæ; (cum & in earum una quavis, per ſe conſiderata, id invenire ſit difficile:) haud etiam fieri poterit, ut ponderum rationes, ſonitibus a ſe factis, perfecte accommodentur: cum &, in eiſdem ipſis tenſionibus, ſpiſſiores & ſubtiliores chordæ, acutiores edant ſonitus. Quinimo (quod potius adhuc eſt) etiamſi quis ea fieri poſſe ſuppoſuerit; atque æqualem item Chordarum longitudinem: pondus tamen majus, majori tenſione, annexæ ſibi chordæ diſtantiam augebit magis, eamque magis denſabit: ut alia proinde hic accedat ſonituum differentia, præter eam quæ eſt a Ponderum ratione. Similia contingunt in ſonitibus pulſando factis; quales in malleis diſciſque, ponderum inæqualium, obſervamus: quique a patellis vacuis pleniſque fiunt. Operoſum utique omnino eſt, in his omnibus, materiarum omnem & figurarum diverſitatem excludere. Verum in Canone (qui dicitur) extenſa Chorda, omnium accuratiſſime & promptiſſime oſtendet nobis conſo-

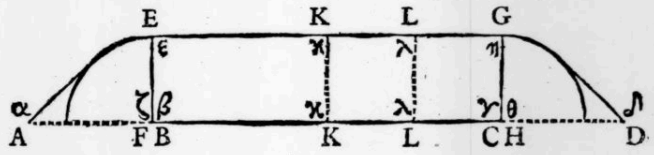
τε τῶν ἐξαπομένων ταῖς χορδαῖς βαρῶν· μὴ ἀφαιρουμένων [h] ἀνωμαλιῶν ἀλλήλαις πολυτάπασι τῶν χορδῶν, ὁπότε καὶ πρὸς [i] αὐτὴν ἐκάστην ὅπως ἔχουσαν εὑρεῖν ἔργον· οὐκέτι δυνατὸν ἔσται τὰς τῶν βαρῶν λόγους ἐφαρμόσαι τοῖς [k] γινομένοις δι' αὐτῶν φθόγγοις· τῶν γὰρ πυκνοτέρας καὶ λεπτοτέρας, ἐν ταῖς αὐταῖς τάσεσιν, ὀξυτέρους φθόγγους ποιεῖν. Πολὺ δὲ ἔτι πρότερον, [l] κἂν τοῦτό τις ὑποθῆται δυνατά, καὶ ἔτι ὃ μήκει τῶν χορδῶν ἴσον· τὸ μᾶλλον βάρει, τῇ πλείονι τάσει, τὴν δ' [m] ἁρτώσιν αὐτῶν χορδῆς διάστασιν αὐξήσει τε καὶ πυκνώσει μᾶλλον· ὥς τε καὶ διὰ τοῦτο συμπίπτειν πυκνὰ, τῶν λόγον τῶν βαρῶν, [n] ἐν τοῖς φθόγγοις ὑπερέχειν. Τὰ παραπλήσια δὲ συμβαίνει, καὶ ἐπὶ τῶν, κατὰ [o] σύγκρασιν, γινομένων φθόγγων· ὁποίους ὀπτῶσι τοὺς διὰ τῶν [p] ἀνίσων σφυρῶν ἢ δίσκων, καὶ τοὺς ἀπὸ τῶν κυμβαλίων κενῶν τε καὶ πεπληρωμένων. Ἐργώδους ὄντος πάνυ τοῦ πράγματος, ἐν ἅπασι τούτοις, τῇ ἐν ταῖς ὕλαις, καὶ τοῖς σχήμασιν, ἀνωμαλίαν. Ἡ [q] δὲ, ὅπου τοῦ καλουμένου Κανόνος, διατεινομένη χορδὴ, δείξει μὲν ἡμῖν τοὺς [r] λόγους τῶν συμφωνιῶν ἀκριβέστερόν τε καὶ προχειροτέρως. Οὐ μὴν ὡς ἔτυχε λαβοῦσα τὰς τάσεις· ἀλλὰ, πρῶτον

h AB. --λίαις *i* C. ἑαυτήν. *k* S. γιγν--. *l* AB. καὶ ταῦτα. *m* AB. ἀρτώσεις αὐτὸ, item GIS. αὐτὸ, ſed manifeſto mendo. *n* AB. deeſt ἐν. *o* GI. σύγκρισιν. *p* S. ἰσοδ--. *q* GI. δ'. *r* S. τόνους.

τον μὲ, μετὰ πινος ἀνακείσεως πρὸς τὴν ἐσομένῳ [f] αὖ ὀκ [t] κατασκευῆς ἀνωμαλίαν. ἔπειτα καὶ τῶν περάτων τὴν προσήκυσαν λαμβανόντων θέσιν· ἵνα τὰ πέρατα τῶν ἐν αὐτοῖς ἀποτελεσμάτων, οἷς ὁ εἶδος τὸ πᾶν μῆκος, οἰκείας τε καὶ δήλας [u] ἔχῃ τὰς ἀρχάς.

nantiarum Rationes. Non autem, tensionem utcunque nacta: sed, post examen primo habitum, nequa foret in ejus constructione inæqualitas: deinde &, extremis debito situ positis; ut Apopsalmatum (*seu Finitorum*) in Canonibus extrema, (quibus tota longitudo deter-

Κανὼν ἁρμονικός. Canon Harmonicus.



αβγδ, εὐθεῖα τοῦ κανόνος. — ABCD, Recta Canonis.
αεηδ, χορδή. — AEGD, Chorda.
αε, ηδ, τὰ δὲ ἐξάμματα. — AE, GD, Ligamenta.
εβ, ηγ, κάθετοι μαγάδος. — EB, GC, Perpendiculares Magadum.
κκ, λλ, μαγάδες κινούμεναι. — KK, LL, Magades mobiles.
βκλγ, κανόνιον. — BKLC, Canonium, seu Regula divisa.

[x] Νοείσθω δὴ Κανὼν ὁ ΧΓΙ τῷ αβγδ εὐθεῖαν· καὶ μαγάδες [y] ποσὶ τοῖς πέρασιν αὐτῶν, πολυτρόπως ἴσας τε καὶ ὁμοίας· [z] σφαιρικὰς, ὡς ἔνι μάλιστα, ποιοῦσαι τὰς [a] ὑπὸ τῆς χορδῆς ὑποφανείας· [b] ἧ τε Β ε, ποσὶ κέντρον τὸ εἰρημένης ὑποφανείας τὸ Ζ· καὶ [c] ἡ Γ η, ποσὶ κέντρον ὁμοίως τὸ Θ. Ληφθέντων [d] τε τῆς ε καὶ η σημείων, ΧΓΙ τὰς διχοτομίας τῆς κυρτῆς ὑποφανῶν· θέσιν ἐχέτως τοιαύτην αἱ μαγάδες, ὥς τε τὰς διὰ τῆς ε καὶ η διχοτομῶν, καὶ τὰς Ζ [e] καὶ Θ

minatur,) apta conspicuaque habeant initia. Intelligatur itaq; Canon secundum ABCD lineam rectam; atque ad ejus extrema Magades *sive Ponticuli*, omnino æquales similesque; sphæricas (quantum fieri licet) superficies, chordis subjectas, exhibentes: ut puta BE, circa dictæ superficiei centrum F; & CG, similiter circa centrum H. Sumptisque punctis E G, in curvarum superficierum bisectionibus; talem habeant Magades positionem, ut quæ a bisectionibus E G cen-

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f K. deest αὖ. *t* C. κατασκευῆς, MSS. Orientales sæpe confundunt ο & β *u* C. ἔχει. *x* S. νοήσω, C. νοείσω. *y* S. deest πρὸς. *z* S. -- καὶ. *a* C. τῆ χορδῆς (quod præfero propter unicam hic chordam,) reliqui τῇ χορδαῖς. *b* ABC. ἥπε. *c* S. deest ἡ. *d* AB. δὲ. *e* ABCS. deest καὶ.

f ABC.

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centriſque F H ducuntur rectæ, puta EBF CGH, perpendiculares ſint ad ABCD rectam. Jam vero ſi, a punctis A D, chordam conterminam diſtendamus, ut AEGD: erit hæc EG ipſi ABCD parallela, (propter æqualem *Magadum ſeu* ponticulorum altitudinem;) atque habebit, in E G punctis, Finitorum *ſeu Apopſalmatum* principia; (quippe in his faciet curvarum ſuperficierum

κέντρων, ἐκβαλλομένας, τυτέστι τὰ f ε̄β̄ζ̄ καὶ τὰ η̄γ̄θ̄, καθέτους ἐπὶ τὸ ἐπίπεδον τὰ α̅β̅γ̅δ̅. Ἐὰν τοίνυν ἀπὸ τῶν α̅ g καὶ δ̅ διατείνωμεν χορδὰς εὐμεγέθεις, ὡς τὴν h α̅ε̅η̅δ̅· ἔσονται δηλονότι αἱ τε ἴσαι k τῇ α̅β̅γ̅δ̅, διὰ τὸ ἴσον l ὕψος ἔχειν τὰς μαγάδας· καὶ λήψεται, κατὰ m τὰ ε̄καὶ η̄ m σημεῖα, τὰς ἀρχὰς τῶν n ἀποψαλμάτων· ἐπ' αὐτῶν γὰρ ποιήσεται τὰς ἐπαφὰς τῶν κυρτῶν ἐπιφανειῶν,

f ABC. ε̄ζ̄ & η̄θ̄γ̄, & ſic infra. g AB. deeſt καὶ. h S. α̅ε̅η̅δ̅. i AB. deeſt τε. k GIS. τῇ. l A B. ἔχειν ὕψος. De ſchemate autem hic expoſito; notandumeſt, Magadas utrinque poſitas (in omnibus codicibus) ſemicirculis deſignari, rectæ α̅β̅γ̅δ̅ (aut huic parallelæ, ultra citrave centra,) inſiſtentibus; ut Cap. 16 lib. 2. quaſi totius Sphæræ ſegmenta exhibentibus, quorum hi ſint ſectiones per axem. Pro quibus ego ſubſtituo circuli quadrantes, Hemiſphærii ſegmenta innuentes, tanquam Ptolemæi ſcopo magis accommodos: tum quia portionum intra ſummitates ε̄η̄ vix ullus compareat uſus; tum quia incommodæ eſſent: Quanquam enim verum ſit ε̄η̄ rectam non niſi in unis punctis ε̄η̄ ſphæras contingere; chorda tamen ε̄η̄, percuſſione mota, ejuſdem ſuperficiei ſphæricæ notabilem aliquam, ſed incertam, partem continget; unde Apopſalmatum puncta ε̄η̄, longitudinem terminantia, reddantur incerta: ſed & ipſæ ε̄ζ̄ η̄θ̄ (ad centra perpendiculares) hac ratione non conſpicientur. Putaverim potius eos ſemicirculos primitus fuiſſe deſtinatos, non ad ſemicirculos in α̅β̅γ̅δ̅ plano deſignandos, ſed ſemicirculos rectæ α̅δ̅ tranſverſim inſiſtentes ad angulos rectos; quorum centra itidem ſint eadem ζ̄θ̄ puncta; & ε̄η̄ ejuſdem κυρτώματος (ſeu curvaturæ) media; unde ε̄β̄ζ̄ η̄γ̄θ̄, (ſi centra ſint infra α̅β̅γ̅δ̅ rectam,) vel ε̄ζ̄ η̄θ̄γ̄ (ſi centra ſint ſupra eandem rectam, demiſſis rectis à ſemicirculi ſeu circularis arcus extremis ad ſubjectum planum α̅β̅γ̅δ̅, nam in hac deſignatione variant codices,) ſint perpendiculares per centra. Sed, ex lib. 2. & 3. videtur potius retinendum fuiſſe Semicirculum; &, per *ſphæricam*, intelligendum *Cylindricam*. Dubium item eſt, an (quod innuit) Canonium, ſit ſeparatim conficiendum, & huic chordæ, ut res poſtulaverit, applicandum (quod dicendum videtur ſi magades fixæ, intra β̅γ̅ puncta continuentur;) an (quod ego potius putaverim) ipſa β̅γ̅ (chordæ ε̄η̄ ſubjecta) ſit hoc Canonium; quod utique, debita ratione diviſum, oſtendat ea notata puncta, quibus recte inſiſtens magadion mobile, imperatas in chorda faciat ſectiones. Huic itaque ſuppoſitioni (ut potiori) accommodavi ſchema noſtrum. In ſequenti vero ſchemate, pro magadis ſuperficie ſphærica, intelligendam exiſtimo magadis oblongæ ſuperficiem cylindricam (plano per axem ſectam) cui (ſecundum planum horizontale) incumbunt (collaterali ſitu poſitæ) chordæ omnes ε̄η̄. Niſi potius illæ omnes chordæ pro una eademq; habeantur (quæ eſt in præcedente ſchemate) alia atq; alia ratione, ſucceſſive, diviſa. m S. τὸ σημεῖον. n S. ἀψαλ-.

o CK.

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ψῶν, διὰ ὃ ΓΔ ε Β Ζ καὶ η Γ θ κεῖται γίνεται. ° Καὶ πρὸς αὐτὸν ἐφαρμόσαντες δὴ τῇ χορδῇ κανόνιον· καὶ μεταλαβόντες P ἐπ' αὐτῆς τὸ Ε η μῆκος, q ἵνα πειρικερότερον ποιώμεθα τὰς παραμετρήσεις· Πρῶτον μὲν r ἐπὶ τὸ γινομένου τοῦ ὅλου μήκους διχοτομίαν, f ὃ ἐστι τὸ Κ, καὶ u ἐπὶ τὴν τῆς ἡμισείας διχοτομίαν, x ὃ ἐστι τὴν Λ, καλαφήσομεν ὑπαγώγια, y τένα εὖ μάλα καὶ z λεπτά, a καὶ νὴ Δία μαγάδια ἕτερα, ὑψηλότερα μὲν ἐκείνων b βραχὺ, ἀπαραλλάκτως δὲ ἔχοντα, θέσεως ἕνεκεν, ἰσότητος, καὶ ὁμοιότητος, c καὶ τῆς μέσης τοῦ κυρτώματος γραμμῆς, ἥτις d ἐστὶ αὐτὸν ἴσην τὴν τοῦ κανόνος διχοτομίαν, e ἢ πάλιν τὴν τῆς f ἡμισείας διχοτομίαν. Ἵνα, ἐὰν μὲν τὸ Ε η τῆς χορδῆς μέρος, ἰσότονον εὑρίσκηται τῷ Κ η· καὶ ἐπὶ τὸ Κ Λ, τῷ Λ η· δῆλον ἡμῖν h αὐτῆς, ᾗ, τὸ κατὰ τὴν σύστασιν ἀπαραλλακτον· Ἐὰν δὲ μή· μεταφέρωμεν τὴν δοκιμασίαν ἐπ' ἄλλο μέρος, ἤτοι χορδῆς ἄλλης, ἕως ἂν τὸ ἀκόλουθον i ἁρμοσθῇ· τυγχάνει, ὃ ἐν τοῖς ὁμοίοις καὶ ἀναλόγοις καὶ ἰσομήκεσι καὶ

contactus, propter EBF CGH perpendiculares.) Ad eandem itaq; *rectam* accommodantes ipſi chordæ Regulam (*ſeu canonium,*) atque in hanc transferentes longitudinem E G, (quo expeditius admenſurationes faciamus:) Primo quidem, ad factam totius longitudinis biſectionem, ut K; deinde, ad ſemiſſis biſectionem, ut L; conſtituamus (*Hypagogia, ſeu*) ſubductoria, tenuiſſima quidem, atque levia; quæ & revera ſint (*Magadia ſive*) Ponticuli alii; prioribus aliquantulo altiores, ſed ſimiliter omnino ſe habentes, tum poſitionis ratione, tum æqualitatis & ſimilitudinis, tum mediæ (curvaminis) lineæ; ſuper regulæ biſectionem ſtatuendæ; itemq; ſuper ſemiſſis biſectionem. Ut nempe, ſi chordæ pars E K comperiatur ipſi K G æquitona; & K L, ipſi L G; hinc nobis conſtet de æquabili chordæ conſtitutione: ſin minus; transferendam eſſe probationem ad aliam chordæ partem, aut etiam ad chordam aliam; uſque dum hæc conditio ſervetur; nempe, ut in ſimilibus, proportionalibus, longitudine æqualibus, ean-

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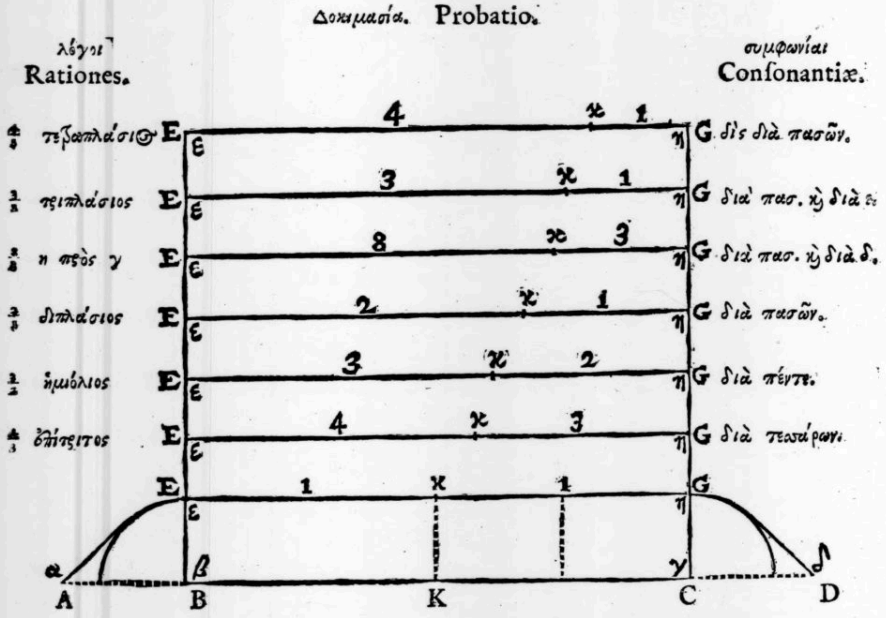
o CK. deeſt καὶ. *p* C. deeſt ἐπ' αὐτῆς. *q* GI. ἵνα. *r* AB. τοῦ. *ſ* C. deeſt οἷον--διχοτομίαν. *t* K. η. *u* GIS. ἔπι, AB. τοῦ. *x* GI. ὡς. *y* GI. πνα. *z* S. λίαν. *a* ABC. ἢ καὶ. *b* C.--χεῖ. *c* K. καὶ. *d* Επ', ſic codex K. reliqui habent ὑπ'. ſed illud præfero, quod & paulo ante præceſſerat: Id nimirum voluiſſe Ptolemæum ſentio, ut Canonium hoc ſit infra chordam, in ipſa AB recta; cui Canonio ita tranſverſim inſiſtant Magadia, ut rectæ à medio curvaturæ ſint ipſis diviſionum punctis perpendiculares. *e* malim καὶ. *f* AB. ἡμίσεως. *g* S. αὖ. *h* CK. αὐτοῖς, AB. αὐτόθεν; GI. αὐτῆς, (quod præfero) ſc. χορδῆς, cujus æquabilitas (τὸ ἀπαραλλακτον) hinc colligitur. *i* C.--θεῖ.

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eandemque tenſionem habentibus, ſit tonus correſpondens. Hoc demum comperto, diviſaque Regula *ſeu Canonio* ſecundum Conſonantiarum rationes jam expoſitas: Inveniemus, (ſubductorii, ad ſingulas ſectiones, applicatione,) auribus comprobatas, quam accuratiſſime, ſonorum correſpondentium differentias. Sumpta enim diſtantia EK talium partium quatuor, qualium KG eſt trium; qui utrinque ſunt ipſarum ſoni, conſonantiam facient Dia-teſſaron; propter rationem ſeſquitertiam. Sumpta vero EK talium trium, qualium

ἐν μίαν ἔχουσι τάσιν, ὁμότονον. Ἐπεὶ δ' τοιούτου καταληφθέντος, καὶ [k] καταδιαιρεθέντος δ' κανονίου τοῖς ἐκκειμένοις τῶν συμφωνιῶν λόγοις· εὑρήσομεν, [l] ἐκ τ, ἐφ' ἑκάτον τμῆμα, δ' μαγαδίου ἀκοαῖς, ὁμολογουμένας ταῖς ἀκοαῖς, ὅτι ὁ [m] ἀκριβέστατον, ὡς [n] τῶν οἰκείων φθόγγων διαφορᾶς. Τοιούτων μὲν δὴ λαμβανομένης τῆς ε κ διαστάσεως τεσσάρων, οἵων ἐστὶν ἡ κ η τριῶν· ἑκάτερον αὐτῶν φθόγγοι ποιήσουσι τὴν διὰ τεσσάρων συμφωνίαν· διὰ τ ἐπίτριτον λόγον. Τοιούτων δὲ λαμβανομένης τ ε κ τριῶν, οἵων ἐστὶν ἡ κ η δύο· ποιήσουσιν



k S. ἀφαιρ--. l S. ἐν τοῖς, C. ἐν τ, sed deleto ἐν. m GI. --τερον. n C. τ.

o AB.

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